

systems to support connectivity features continues to rise.

Focus on safety and automotive regulations. Governments worldwide are imposing stricter regulations to ensure the safety of vehicles, especially in areas like battery safety in EVs. This has led to a push for innovation within automotive electronics.

Autonomous vehicles. The development of autonomous vehicles introduces a whole new array of sensors and electronics, including lidars, advanced image sensors, radars, and ultrasonics. The level of autonomy directly impacts the number and complexity of sensors required.

Demand for connectivity and infotainment features. Consumers increasingly seek connected and infotainment features in their vehicles, further driving the demand for automotive electronics.

The future of automotive technology hinges on the perfect synergy between software precision and essential hardware components. While software excellence is crucial, sensors like lidar, advanced image sensors, radar, and ultrasonics play a vital supporting role.

As vehicles advance from autonomy levels one to five, the demand for sensors skyrockets. This surge is a significant growth driver, aligning with the increasing demand for connectivity and advanced infotainment in the automotive sector.

Challenges in the automotive electronics industry

While opportunities abound, the automotive electronics industry also faces significant challenges that must be addressed. In the realm of automotive electronics, several challenges have come to the forefront.

First and foremost, the heightened connectivity of vehicles has given rise to mounting concerns regarding cybersecurity. The industry is grappling with the rising incidence of software threats, attacks, and recalls stemming from security vulnerabilities, posing a significant hurdle to progress. Furthermore, the intricacies surrounding the design and manufacture of scalable automotive electronics have proven formidable.

Establishing a resilient supply chain has become imperative to ensure uninterrupted production, which demands meticulous attention. Moreover, the escalating costs associated with raw materials, logistics, and the establishment of manufacturing hubs have cast a shadow on the cost-effectiveness of electronic components, warranting careful consideration within the industry.

Challenges that surfaced towards the latter part of 2021 and extended into 2022 underscored the neces-

Stay Powered on the Go: Protected USB port on the motorcycle with the LDP01-28AYD2-AQ from Diotec

As spring arrives and the days get longer, many motorcyclists take their bikes out of storage and hit the road. A smartphone is a must-have accessory, as there are apps available to help bikers navigate, find the best and cheapest fuel stops, and provide weather information for the route. Even in the event of an accident, they can send an emergency call with location information to the nearest rescue coordination center and eliminate the need to search for an SOS emergency call center, which is unfortunately too far away in most cases.



On longer journeys, the smartphone battery may run out at some point. You can easily charge your smartphone using the USB port in the cockpit. Like all electronics in a vehicle, it must be protected against load-dump events. These are voltage peaks on the board mains, occurring typically in the case of a jump start. When the supplying battery is disconnected under load, the jumper cables act like an ignition coil. Here, the Load Dump Protector LDP01-28AYD2-AQ by Diotec Semiconductor comes into play. According to ISO 16750-2 load dump curves, it offers protection and clamps its peaks to safe levels. The device is fully AEC-Q101 qualified and comes in the powerful D²PAK case outline.

FOR MORE INFO



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